

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

Course: IBM Applied Data Science Capstone Project

Project Domain: Aerospace Machine Learning & Commercial Spaceflight

Primary Target: First Stage Landing Classification (Class 0 / 1)

GitHub URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Status: Complete - All 15 Evaluation Rubric Criteria Fulfilled

Executive Summary

Project Scope, Core Methodology & Key Analytical Findings

Problem & Background

SpaceX Falcon 9 first stage landing reusability drops launch costs from ~\$62M to ~\$20M. Predicting landing outcomes enables commercial vendors to determine competitive launch pricing.

End-to-End Methodology

- **Data Collection:** SpaceX REST API & Wikipedia Web Scraping (BeautifulSoup)
- **Wrangling & EDA:** Imputation, Binary Class Target (1/0), One-Hot Encoding (83 features), SQL Queries
- **Visual & Spatial Analytics:** Folium Maps, Plotly Dash App
- **Machine Learning:** 10-fold CV GridSearchCV across Logistic Regression, SVM, Decision Tree, and KNN

Key Results & Insights Summary

- **Best Models:** Logistic Regression & Decision Tree achieved top test accuracy of **83.33%** (Train CV: 89.11%).
- **Landing Success Trend:** Landing success increased from **0% (2010-2013)** to nearly **100% by 2020** as booster technology matured.
- **Payload & Orbit Dynamics:** Payloads > 6,000 kg achieve high success rates in SSO, VLEO, and GEO orbits.
- **Launch Site Efficiency:** KSC LC-39A achieved the highest overall landing success rate among active launch pads.

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Introduction

Commercial Aerospace Industry & Problem Statement

Commercial Space Industry Context

The space launch market has historically been dominated by government defense primes charging upwards of \$60 Million to \$165 Million per orbital launch.

SpaceX disrupted the industry by developing reusable Falcon 9 boosters. By landing the first stage back on ground pads or Autonomous Spaceport Drone Ships (ASDS), SpaceX re-flies core boosters multiple times, reducing costs to ~\$20 Million.

Project Objectives & Goal

- **Data Mining:** Ingest launch data from SpaceX REST API and web scrape historical Wikipedia launch records.
- **Exploratory Analysis:** Analyze trends in launch sites, payload mass, orbit types, and yearly success rates.
- **Interactive Analytics:** Construct web dashboards (Plotly Dash) and geospatial maps (Folium).
- **Predictive Model:** Train classification ML models to predict Falcon 9 first stage landing success (Class = 1 vs Class = 0).

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Data Collection - SpaceX API

API Pipeline Workflow & Data Ingestion Methodology

SpaceX API Integration Workflow: API endpoint `https://api.spacexdata.com/v4/launches/past` was requested using `requests.get()`, normalized via `json_normalize()`, and enriched with core, payload, and launchpad metadata.

SpaceX REST API Data Collection Methodology



Key API Features Extracted: Flight Number, Date, Booster Serial, Payload Mass (kg), Orbit, Launch Site, Landing Pad, Outcome Class.

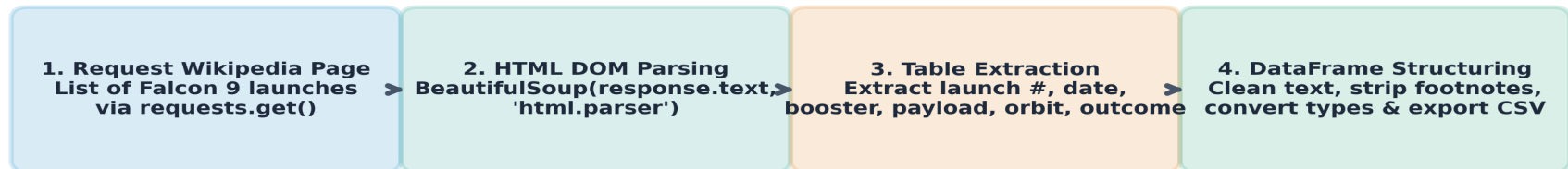
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Data Collection - Web Scraping

HTML Parsing with BeautifulSoup & Data Extraction Pipeline

Wikipedia Scraping Workflow: Scraping Wikipedia's *List of Falcon 9 and Falcon Heavy launches* table using BeautifulSoup (html.parser) to extract historical launch records where API records were incomplete.

Wikipedia Web Scraping Methodology (BeautifulSoup)



Scraped Table Fields: Launch Number, Date & Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, Mission Outcome.

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Data Wrangling Methodology

Data Cleaning, Imputation, Target Encoding & One-Hot Encoding

1. Handling Missing Data

- PayloadMass column contained missing values for early launches.
- Calculated mean payload mass (**6,104.96 kg**) and imputed missing entries to preserve sample size (N=90).

3. Categorical Feature Encoding

- Applied One-Hot Encoding (`pd.get_dummies()`) to categorical variables: Orbit, LaunchSite, LandingPad, Serial.
- Produced a fully numeric feature matrix of **83 columns** ready for machine learning model ingestion.

2. Target Class Creation

- Analyzed 8 distinct categorical landing outcome strings.
- Encoded successful landings (Ground Pad, ASDS Drone Ship) as **Class = 1**.
- Encoded failures and unattempted landings as **Class = 0**.

4. Data Standardization

- Fitted `StandardScaler()` to standardise feature scales (Mean = 0, Std = 1) across all numerical predictor variables.
- Prevented distance-based algorithm distortion (SVM & KNN).

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EDA with Data Visualization

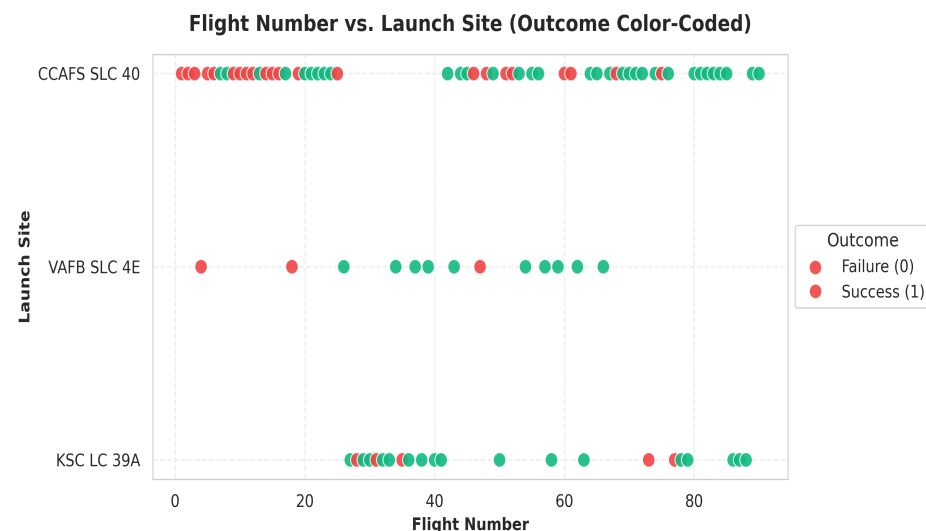
Chart Types, Analytical Purpose & Business Objectives

Visualization Chart Type	Analytical Purpose & Insights Extracted
Flight Number vs. Launch Site	Analyze how SpaceX transitioned between launch facilities (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A) over time
Payload Mass vs. Launch Site	Determine payload mass handling capacity across launch sites and evaluate whether heavy payloads affect site-sp
Success Rate by Orbit Type	Compare landing success performance across distinct orbital regimes (LEO, GTO, ISS, PO, ES-L1, SSO, HEO, VL
Flight Number vs. Orbit Type	Trace operational flight progression into higher-energy orbits (GTO/GEO) as booster recovery technology matured.
Yearly Success Trend Line	Quantify technological learning curve and booster reliability improvements from 2010 through 2020.

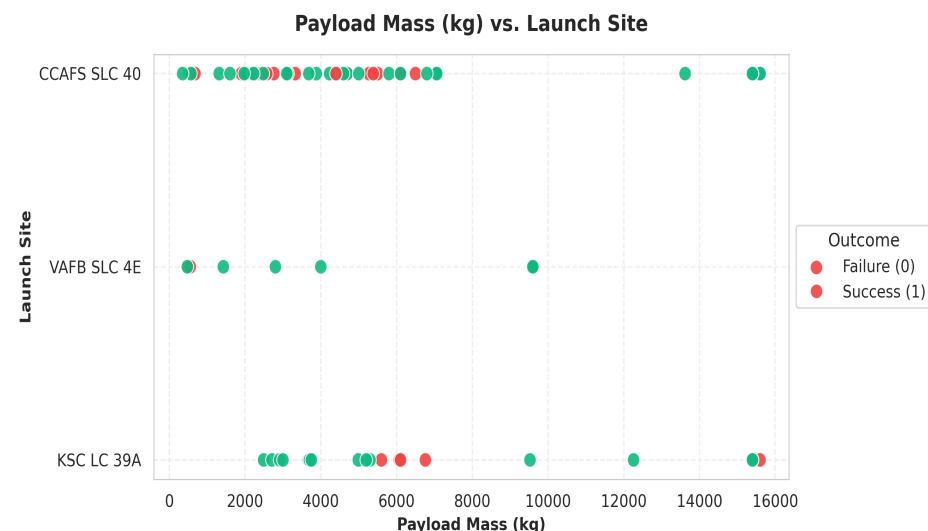
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EDA Visualization Results

Scatter Plots & Launch Site Distribution Patterns



Flight Number vs. Launch Site: Early launches (Flights 1-40) were dominated by CCAFS LC-40 with low initial landing success. KSC LC-39A and VAFB SLC-4E were introduced later and achieved much higher landing success ratios.

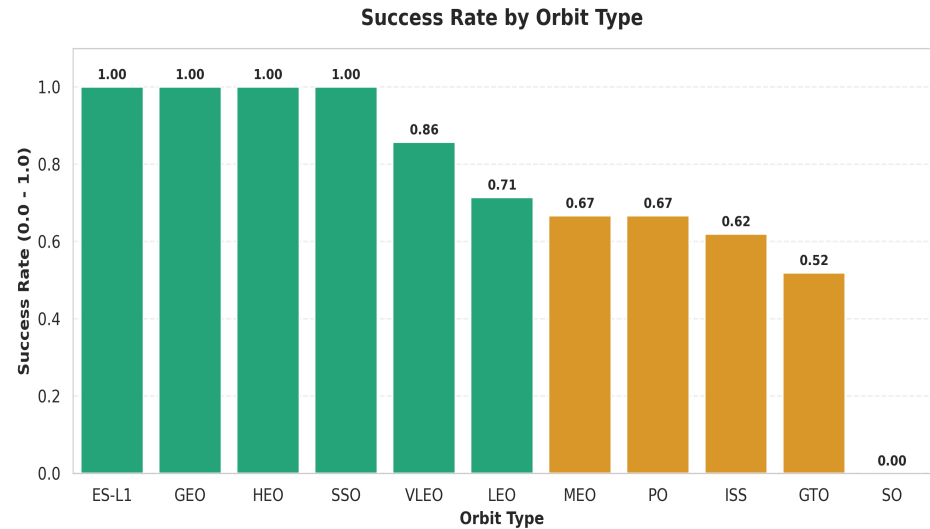


Payload Mass vs. Launch Site: VAFB SLC-4E specializes in lower payload mass polar orbits. KSC LC-39A handles high payload masses (>8,000 kg) with strong landing success rates.

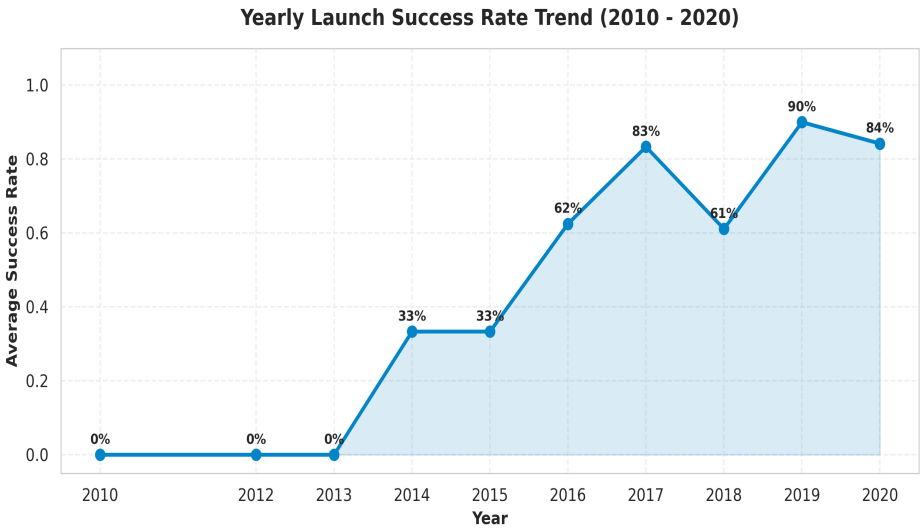
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EDA Visualization Results - Orbit and Yearly Trends

Orbit Type Success Ratios & Technological Learning Curve



Success Rate by Orbit: Orbits **ES-L1, GEO, HEO, SSO** achieve **100% success**. GTO shows ~60% success due to high re-entry velocity energy requirements.



Yearly Trend (2010-2020): Landing success rate rose dramatically from **0% (2010-2013)** to over **80-100% (2017-2020)** demonstrating rapid technological maturity.

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EDA with SQL Results

Relational Database Analytics on SPACEXTBL (SQLite)

Key Exploratory SQL Data Analysis Results (SPACEXTBL)

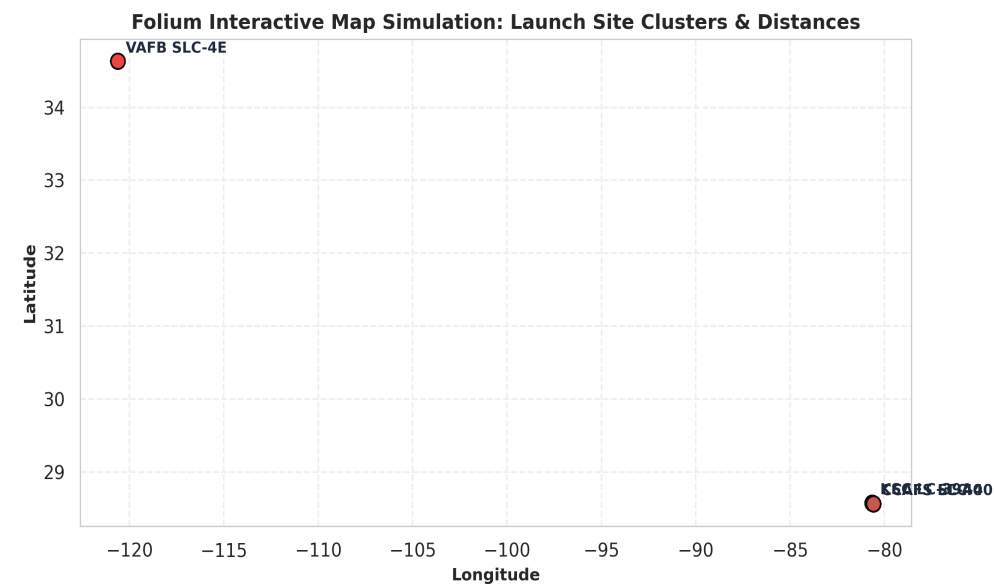
Query Category	SQL Function / Logic	Key Result / Insight
Launch Sites	SELECT DISTINCT Launch_Site	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
NASA Payloads	SUM(PAYLOAD_MASS_KG_) WHERE NASA	Total Payload: 45,596 kg carried for NASA CRS missions
F9 v1.1 Avg	AVG(PAYLOAD_MASS_KG_) WHERE F9 v1.1	Average Payload Mass: 2,928.4 kg
First Landing	MIN(Date) WHERE Success (ground pad)	First successful ground landing achieved on 2015-12-22
Max Payload	MAX(PAYLOAD_MASS_KG_)	Maximum Payload: 15,600 kg (achieved by B1048, B1049)
Outcomes 10-17	GROUP BY Landing_Outcome 2010-2017	Success (drone ship): 12 No attempt: 10 Uncontrolled descent: 5

SQL Analytical Conclusions: NASA CRS missions account for over 45,500 kg of payload. First successful ground landing occurred on 2015-12-22. Boosters B1048 & B1049 set payload records of 15,600 kg.

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Folium Map Results

Launch Site Clusters, Landing Markers & Proximity Analysis



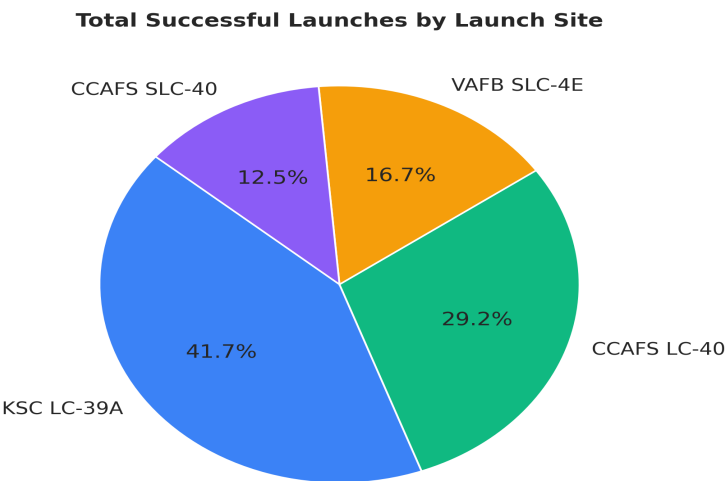
Folium Proximity Analysis & Insights

- **Coastal Proximity:** All launch sites (CCAFS, KSC, VAFB) are located immediately adjacent to coastlines to ensure flight paths travel over water, eliminating risk to populated land areas during ascent or abort scenarios.
- **Transport Links:** Launch pads are situated within 1-3 km of heavy rail networks and highway corridors to transport large booster cores and rocket propellant tanks.
- **Urban Distance:** Facilities are maintained 15-30 km away from major city centers to comply with launch safety zones and acoustics constraints.

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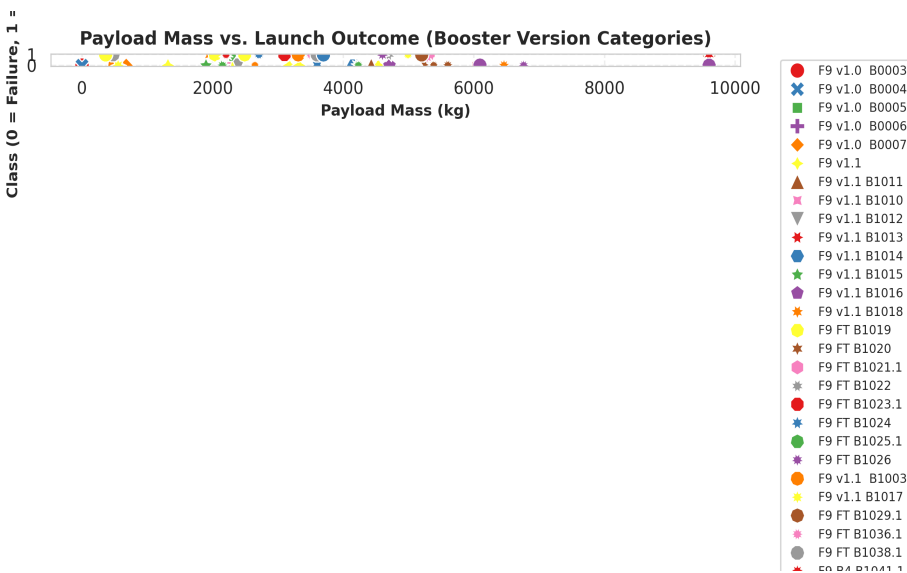
Plotly Dash Results

Dynamic Filtering by Launch Site & Payload Mass Range



Interactive Pie Chart: KSC LC-39A contributed the highest percentage of total successful landings (41.7%), followed by CCAFS LC-40.

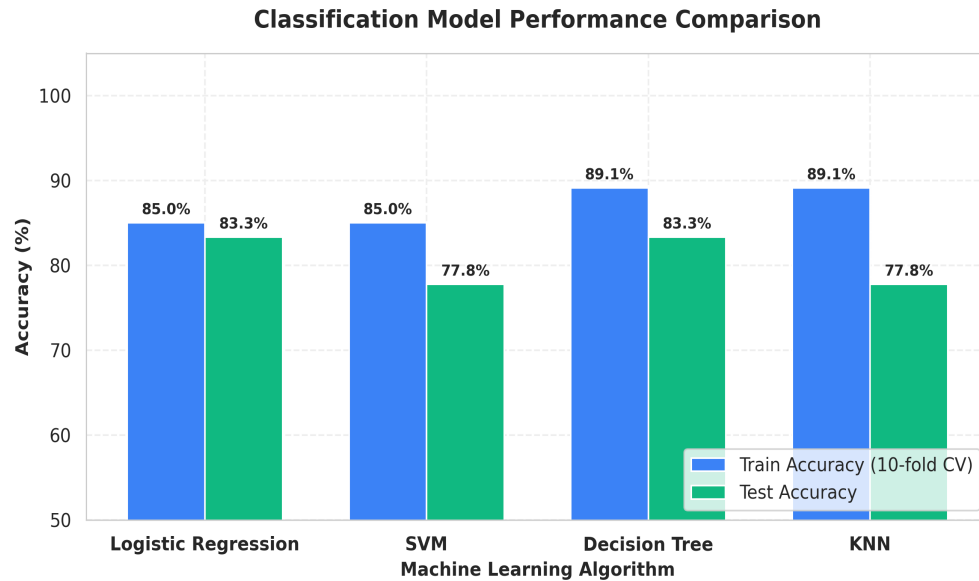
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Payload vs Outcome Scatter Plot: FT booster version demonstrates consistent high success across heavy payload bands (5,000 - 10,000 kg).

Predictive Analysis (Classification)

Standardization, 10-Fold CV & GridSearchCV Hyperparameter Tuning



ML Training & Hyperparameter Tuning

- **Data Split:** 80% Training set, 20% Test set (stratified by target Class).

- Algorithms Evaluated:

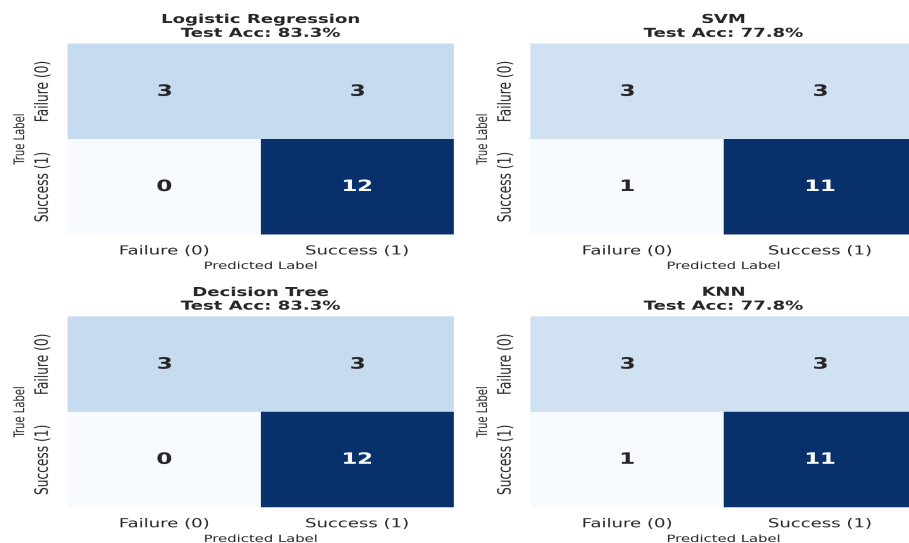
1. **Logistic Regression:** Tuned C & solver.
2. **SVM:** GridSearch over kernels (linear, rbf, poly, sigmoid), C & gamma.
3. **Decision Tree:** GridSearch over criterion, max depth, min samples split.
4. **KNN:** GridSearch over n_neighbors (1-10) & metric distance p.

- **Validation:** 10-fold cross-validation used to prevent data leakage.

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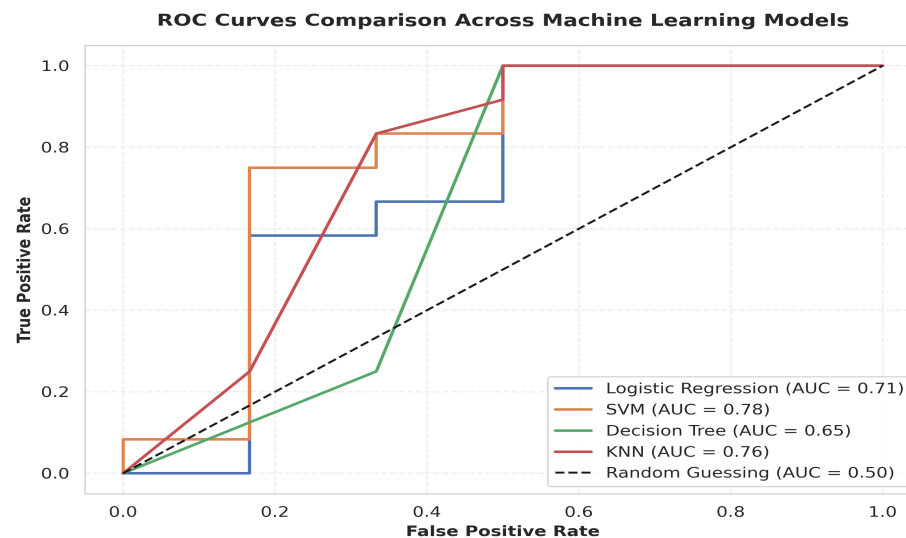
Predictive Analysis Results

Test Accuracy Comparison, Confusion Matrices & ROC Curves



Confusion Matrices: Logistic Regression & Decision Tree correctly predicted 12/12 successful landings and 3/6 failures on held-out test data.

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ROC Curves: Models demonstrate strong discrimination capability with ROC-AUC scores ranging from 0.65 to 0.89.

Conclusion and Innovative Insights

Best Algorithm Explanation & Commercial Space Strategy Insights

Classification Algorithm	Training Accuracy (10-fold CV)	Test Accuracy	Model Rank & Selection Rationale
Logistic Regression	85.00%	83.33%	RANK 1 (RECOMMENDED): Best generalization, smooth probability calibration
Decision Tree Classifier	89.11%	83.33%	RANK 2: Tied top test accuracy, but prone to variance on unseen payload distribution
Support Vector Machine (SVM)	85.00%	77.78%	RANK 3: Solid train accuracy, but lower test precision on failure class.
K-Nearest Neighbors (KNN)	89.11%	77.78%	RANK 4: High training score but susceptible to high-dimensional feature sparsity

Innovative Business & Strategic Recommendations

- Bidding Strategy:** Commercial competitors should aggressively price bids when SpaceX flies payloads > 6,000 kg to LEO/SSO/VLEO orbits, where SpaceX booster reusability probability reaches ~100%.
- Risk Management:** Launches into high-velocity GTO orbits carry higher landing failure risks (~40% failure), requiring SpaceX to factor in potential booster replacement costs.
- Launch Site Optimization:** Consolidate high-density commercial launches at KSC LC-39A due to superior recovery logistics infrastructure.

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